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symbiosis LABS

Turning mitochondria into a programmable interface for human health and performance.

Less than 3% of global disease burden comes from single-gene mutations. The other 97%, including neurodegenerative, muscular, and other age-associated diseases, involve tens or hundreds of genes acting together, or emerge from lifestyle and environmental factors. These conditions live upstream of any single gene, in higher-order cellular states. To solve them and expand human performance, we must go beyond targeting singular genes and proteins. We must modulate cellular control nodes through an interface that senses, integrates, and modulates these states at single-cell resolution and at a full-body scale.

Nature has already evolved one, our mitochondria.

Mitochondria are not just the powerhouse of the cell. **They are the control room**. Each is a tiny cellular machine living inside our cells, generating 80% of cellular ATP, setting the cytoplasmic calcium that gates synaptic plasticity and muscle contraction, producing the reactive oxygen species that tell cells to adapt, and making the final call for cell death. They also transfer naturally between cells in muscle, brain, blood, and tumors.

We cannot engineer human mitochondria today. Their 16,569-base genome is 180,000 times smaller than the nuclear genome and is enclosed by two membranes that cannot be punctured without killing the cell. Symbiosis Labs, an applied research lab, was founded to close this gap and install new genetic information into human mitochondria. It is turning mitochondria into a programmable interface for human health and performance. By developing tools to control mitochondria, we'll offer a new paradigm for interfacing with our cells, not through individual gene edits, but through modulation of higher-order signals, pushing the limits of human health and performance.

The Challenge

We cannot add new genes to our mitochondria. While there are ample tools to modify the nuclear genome, the human mitochondrial genome remains out of reach. Existing approaches allow editing of specific base pairs or cleavage of certain mtDNA molecules, but fall short of inserting new genetic information.

Two technical problems limit our ability to engineer mitochondria. Mitochondria have integrated so deeply into our cells that they have lost many functions of free-living bacteria. This includes the ability to import nucleic acids: no plasmid conjugation, no known viruses targeting them, and no machinery for transporting DNA from the environment. Commonly used DNA delivery approaches for mammalian nuclei and bacteria are therefore obsolete. Other methods for DNA delivery harm membrane integrity, which mitochondria do not tolerate. Mitochondria have an inner membrane, which uses a proton gradient to generate ATP, and an outer membrane, which coordinates with the cytoplasm. Any attempt to deliver DNA into mitochondria can disrupt either membrane, causing protons in the intermembrane space to flood into the matrix and impair ATP production. This is called depolarization. Puncturing the outer membrane is worse- even a pore the diameter of DNA enables cytochrome C to leak into the cytoplasm and trigger cell death.

Technical Approach

Viruses are nature's most robust delivery systems for DNA and RNA in living systems. Although mitochondria have a bacterial origin, their hundreds of millions of years spent secluded in the intracellular environment have left them without naturally occurring viruses that infect them. Thus, we want to re-engineer viral targeting systems to infect mitochondria. We are pursuing this through two orthogonal methods: modifying mitochondria to be susceptible to naturally occurring viruses, and modifying viruses to target native mitochondrial outer membrane proteins.

In preliminary work, we have shown (I) robust engineering of our candidate phage, (II) encapsulation of phage in fusogenic liposome systems, (III) the ability to traffic its receptor to mammalian mitochondria, and (IV) naked mitochondria with trafficked phage receptor having the full phage genome injected into them. Together, these results **establish the feasibility of the core steps required for this platform** and position us to assemble them into the first programmable system for introducing DNA into mitochondria.

Translation: From Rare Diseases to a Programmable Interface

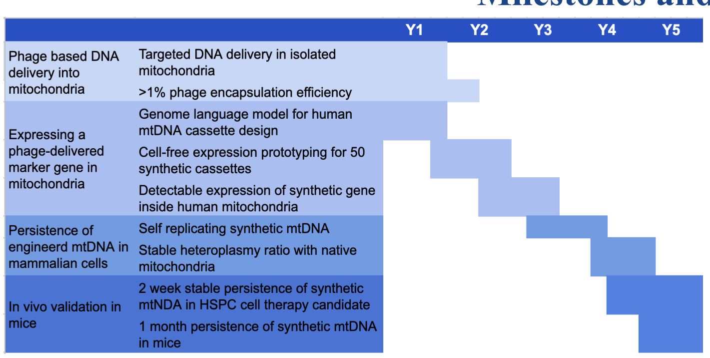
The design of a programmable mitochondrial interface is staged. The first five years of this program deliver curative treatments for primary mitochondrial diseases. The second five years extend the platform to a broader range of diseases and to human performance. Programmable mitochondria are a foundation layer for interfacing with the human body at the molecular level, **with single-cellular precision and whole-body scale**.

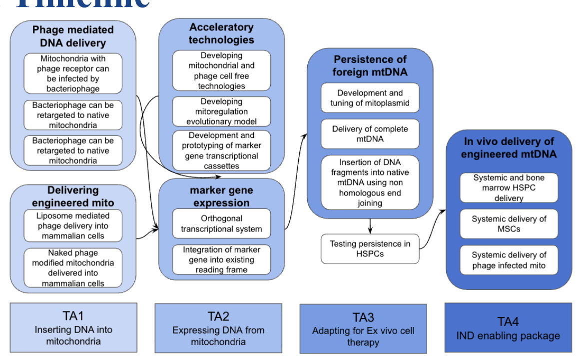
Primary mitochondrial diseases are a group of rare, incurable, and typically childhood-onset disorders. Roughly 1 in 4,300 people carry a pathogenic mtDNA variant. Affected children face progressive neurological decline, organ failure, and early mortality, with no therapies available. The technology generated by this program is intended to directly enable compassionate-use trials for patients within the first five-year window.

Beyond primary mitochondrial disease, **mitochondrial dysfunction is already recognized as a driver** of a much larger set of conditions, including neurodegeneration, hematopoietic disease, muscular disease, and other age-associated pathologies. CNS delivery barriers put neurodegenerative indications such as Parkinson's out of near-term reach. However, in hematopoietic diseases, cell therapies developed for primary mitochondrial disease apply directly to disorders such as myelodysplastic syndromes. In muscular disease, the same therapies extend to indications such as sarcopenia with minimal adaptation.

In a broader sense, mitochondrial output, including ATP supply, reactive oxygen species handling, and calcium signaling, **governs human performance**- including synaptic plasticity, resistance to radiation and microgravity, cognitive performance under load, thermogenesis, hypoxia tolerance, muscle recovery, and more. For example, in space health, mitochondria are the weakest link: radiation damages mtDNA at higher rates than nuclear DNA, and microgravity disrupts mitochondrial dynamics and metabolism. A programmable mitochondrial interface addresses these directly: genome-repair cassettes that maintain mtDNA integrity under chronic radiation, buffering of reactive oxygen species, preservation of mitochondrial metabolism, and even organelle-intrinsic radiation shielding modeled on extremophile biology. Over a longer time horizon, **mitochondrial engineering converts some of these physiological limits into design parameters**.

Milestones and Timeline





These four aims are deliberately chosen to clear the minimum technical bar required to transition mitochondria into a programmable genetic chassis. **TA1: delivery**, addresses the single most-cited reason mitochondrial engineering has stalled for decades and, on its own, enables ex vivo augmentation strategies that several mitochondrial-disease companies could adopt immediately. **TA2: expression**, extends this from a delivery vehicle to a programmable interface, enabling arbitrary transgenes to be expressed in mitochondria in a tunable manner, which is a prerequisite for any therapeutic, performance, or longevity payload. **TA3: persistence,** determines whether engineered mtDNA behaves as a stable genetic element in dividing cells, the gating question for any in vivo or HSPC-based product. **TA4: *in vivo* function** demonstrates that engineered mtDNA is durable in an organismal context, which is the de-risking step industry needs before committing to clinical development.

Risks and Mitigation Strategies

The four technical areas above each carry known risks for which we have planned mitigations, and unknown risks for which we have planned fallback paths:

TA	Type	Risk	Mitigation
TA1	Known	Retargeting phage to mitochondria is hard; de novo mitochondrial phage targeting is high-risk.	"Disguise" mitochondria by presenting a native phage receptor on mitochondria; pursue protein-design retargeting in parallel as a path.
		No optimized method exists for cytosolic phage delivery; unencapsulated phage is unstable/cleared and can be endocytosed then degraded.	Encapsulate in lipid particles designed for endocytosis + endosomal escape; optimize composition/size for delivery.
		Delivery efficiency of liposome-based phage delivery may be insufficient.	High-throughput microfluidic formulation + DoE; test multiple encapsulation routes, and non liposome methods as fallback.
		Phage injection may induce excessive mitochondrial membrane depolarization; failure to recover $\Delta\Psi m$ is a critical failure mode.	Pharmacologically inhibit collapse after initial permeabilization; if not recoverable, switch to a different injection system (e.g., contractile injection).
		Our candidate is challenging to engineer due to resistance to endonucleases (e.g., CRISPR-Cas9).	Use a discovered permissive insertion site plus Cas13-based selection to install payload cassettes with limited fitness penalty.
	Unknown	Phage may not function properly within cytoplasmic context	Persue ex-vitro phage infection in isolated mitochondria and deliver phage-infected mitochondria via mitochondrial augmentation (MAT).
TA2	Known	Mitochondrial transcription–replication logic unknown and non-modular, limiting predictable expression engineering.	Build an orthogonal expression layer and keep an integration into existing reading frame as fallback.
	Known	Orthogonal mitochondrial expression may yield no detectable/usable expression.	Pivot to inferred native regulatory motifs via comparative/co-evolutionary analysis; final fallback is fusion to a native mt gene (higher interference risk).
TA3	Known	Large inserts/duplications likely won't persist long-term; engineered DNA can be at replicative/segregative disadvantage vs native mtDNA.	Use minimal "mitoplasmid" backbones (only essential regulatory elements) rather than bulky modified full genomes.
	Known	Persistence/segregation dynamics may be heterogeneous; bulk assays may miss subclonal loss/expansion.	Complement bulk qPCR/luminescence with single-cell readouts (high-throughput imaging and/or flow cytometry of marker expression).
	Known	Mitoplasmids may fail to persist or be rapidly eliminated by mitochondrial quality control.	Time-boxed pivot (≤6 months): switch to targeted integration using mito-endonucleases + ligase/NHEJ-like repair with intrinsic selection against unedited mtDNA.
	Unknown	Mammalian mtDNA cannot be stably propagated in E. coli, creating unpredictable phage fitness costs for mtDNA-derived payloads.	Use mtDNA fragments (not full genomes) and bootstrap phage assembly using a cell-free system.
	Unknown	Persistence may be highly context-dependent across mito–nuclear genetic backgrounds (risk of non-transferability).	maintain the same genetic background in tests, optionally expand to testing and optimizing for multiple backgrounds
TA4	Known	Persistently expressing engineered mtDNA in non-hematopoietic/endogenous cell types is the higher-risk in vivo step.	Adapt by switching cell types (e.g., MSCs), changing administration routes, and/or using host genetic modifications expected to enable persistence/integration.
	Unknown	In vivo biodistribution and optimal routes/tissues for direct mitochondrial injection are uncertain and fast-changing.	Use the latest biodistribution evidence to choose routes; empirically test multiple routes/timepoints and proceed with whichever achieves persistence.

Summary

Symbiosis Labs is leading a moonshot program aiming to convert mitochondria from an inaccessible organelle into a programmable interface. The program is executed in collaboration with Imperial College London (lipid nanoparticle development and formulation for phage delivery) and Minovia Therapeutics (preclinical and clinical HSPC-mitochondrial augmentation therapies).

By the end of five years, we will deliver the full toolkit and transition mitochondria from an inaccessible organelle into a programmable interface with cells. This mission transcends any single research effort. **We will democratize the interface** by hosting visiting scholars for training, releasing systematic toolkits with validated protocols, and driving integration across every domain of biology that would benefit from a cell-level interface. The ability to manipulate the nuclear genome gave biology a shared engineering toolkit; researchers in genetics, immunology, and developmental biology developed shared methods, vocabulary, and perspectives on gene function, bridging the fields into a unified understanding of our genomes.

Since mitochondria are the control room of our cells, **this program will unify fields** such as metabolism, neurobiology, oncology, and regenerative medicine around shared tools and perspectives, **illuminating the dynamic state and behavior of cells.**

Selected Citations and Supplementary References

Mitochondria as multifaceted regulators of cellular state. Mitochondria are recognized as central regulators of cellular physiology beyond ATP production, with established roles in redox balance, calcium buffering, innate immune activation, programmed cell death, hormone and cofactor synthesis, and metabolic signaling (Nunnari & Suomalainen 2012, *Cell*; Bock & Tait 2020, *Nat. Rev. Mol. Cell Biol.*; Picard & Shirihai 2022, *Cell Metab.*; Mills, Kelly & O'Neill 2017, *Nat. Immunol.*; Monzel, Enríquez & Picard 2023, *Nat. Metab.*). These functions are tightly interdependent. Disruption of electron transport destabilizes redox balance, damages DNA, impairs mitophagy, and triggers inflammation, with effects that propagate across tissues and organisms.

Mitochondrial dysfunction is upstream of chronic and age-related disease. Mitochondrial failures contribute to a broad set of clinically significant conditions that are not primarily monogenic, including Parkinson's disease, amyotrophic lateral sclerosis, type 2 diabetes, fatty liver disease, infertility, and therapy-resistant cancer (Murphy & Hartley 2018, *Nat. Rev. Drug Discov.*; Bose & Beal 2016, *J. Neurochem.*). Mitochondrial decline is also a hallmark of aging, driven by accumulated mtDNA mutations, impaired organelle recycling, and persistent oxidative stress (López-Otín et al. 2013, *Cell*; López-Otín et al. 2023, *Cell*; Bilen et al. 2022, *Curr. Opin. Physiol.*). In these conditions, the underlying lesion is rarely a single gene, and modulation at the level of organelle state offers a route that gene-by-gene approaches do not.

Primary mitochondrial disease prevalence and unmet need. Primary mitochondrial diseases (PMDs), caused by inherited mutations in mtDNA or nuclear genes, affect roughly 1 in 4,300 births and typically result in multi-organ failure with no adequate treatments available (Gorman et al. 2015, *Ann. Neurol.*; Russell, Gorman, Lightowlers & Turnbull 2020, *Cell*, Schofield et al. 2025, *Value Health*).

Natural intercellular mitochondrial transfer. Mitochondria move between cells in tissue repair, immune signaling, and the tumor microenvironment, and have been documented in muscle, neurons, hematopoietic lineages, and across tumor stroma (Liu et al. 2021, *Signal Transduct. Target. Ther.*; Borcherding & Brestoff 2023, *Nature*). This natural transmissibility positions mitochondria as an unusual engineering substrate: they carry their own genome, depend on nuclear-encoded partners, and physically move within and between cells.

Current state of mitochondrial genome engineering: edit but not insert. Available tools allow targeted editing or degradation of existing mtDNA but cannot insert new genetic information. Mitochondrially-targeted zinc finger nucleases (mitoZFNs) selectively degrade mtDNA carrying large-scale deletions or point mutations (Gammage et al. 2014, *EMBO Mol. Med.*). MitoTALENs have eliminated mutant mtDNA in MELAS-derived iPSCs (Yang et al. 2018, *Protein Cell*). DdCBE base editors achieve mitochondrial base editing in human embryos and in adult mouse tissue via AAV delivery (Chen et al. 2022, *Cell Discov.*; Silva-Pinheiro et al. 2022, *Nat. Commun.*). Heteroplasmy manipulation remains technically unruly. Field reviews note that the mitochondrial revolution "may not be CRISPR-ized" by these approaches alone (Gammage, Moraes & Minczuk 2018, *Trends Genet.*; Jackson et al. 2020, *Trends Mol. Med.*; Silva-Pinheiro & Minczuk 2022, *Nat. Rev. Genet.*).

Clinical pipeline and regulatory precedent. Mitochondrial donation in IVF is approved in the United Kingdom for at-risk patients (Gorman, Grady & Turnbull 2015, *N. Engl. J. Med.*; Craven, Murphy & Turnbull 2020, *Emerg. Top. Life Sci.*). Mitochondrial transplantation has entered early human testing for ischemia-reperfusion injury (Lightowlers, Chrzanowska-Lightowlers & Russell 2020, *EMBO Rep.*). Mitochondrially-targeted small molecules and metabolic agents are advancing in parallel, including elamipretide (Tung et al. 2025, *Int. J. Mol. Sci.*), MitoQ (Lowes et al. 2008, *Free Radic. Biol. Med.*; Mason et al. 2022, *Diabetes Obes. Metab.*), and the mitophagy enhancer urolithin A. These clinical precedents establish regulatory pathways for mitochondrial interventions while operating downstream of the genetic layer that the present program addresses.